

NARYAN-MAR, Russian Federation

WMO: 232050

Lat: 67.6249N

Lon: 53.0175E

Elev: 12

StdP: 101.18

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.0	-34.8	-40.7	0.1	-37.8	-37.5	0.1	-34.8	9.4	-8.9	8.8	-9.3	1.0	240	0.783

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	26.1	18.4	23.2	16.8	20.7	15.4	19.4	24.5	17.8	22.2	16.0	19.8	3.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.4	12.5	22.0	15.8	11.2	20.2	14.1	10.1	18.5	55.3	24.7	50.2	22.5	45.1	20.0	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
8.5	7.6	6.9	DB	-40.4	28.9	3.6	2.8	-43.1	30.9	-45.2	32.6	-47.2	34.1	-49.9	36.1	
			WB	-40.6	20.8	3.5	1.9	-43.1	22.2	-45.2	23.3	-47.1	24.3	-49.7	25.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.6	-17.1	-17.0	-11.3	-5.2	1.1	8.5	13.5	10.6	6.1	-0.3	-8.8	-12.3
	DBStd	12.40	9.34	9.92	7.79	6.29	5.49	5.45	5.15	3.75	3.52	4.80	8.01	9.07
	HDD10.0	4846	839	756	662	456	281	93	23	37	125	321	563	692
	HDD18.3	7662	1097	989	920	706	533	299	168	241	367	579	813	951
	CDD10.0	248	0	0	0	0	6	48	130	56	8	0	0	0
	CDD18.3	23	0	0	0	0	0	4	17	2	0	0	0	0
	CDH23.3	222	0	0	0	0	2	46	162	13	0	0	0	0
	CDH26.7	49	0	0	0	0	0	9	38	2	0	0	0	0
Wind	WSAvg	3.3	3.3	3.3	3.4	3.5	3.5	3.6	3.1	3.1	3.2	3.3	3.2	3.4
Precipitation	PrecAvg	469	28	22	24	26	33	44	49	64	55	47	38	34
	PrecMax	650	54	60	50	65	86	134	125	136	141	86	99	68
	PrecMin	281	9	2	5	4	1	12	15	12	4	17	14	12
	PrecStd	94	10	11	10	14	18	27	27	28	29	16	16	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.6	1.2	3.3	9.5	20.9	27.9	30.0	25.1	18.4	11.2	3.7	1.7
		MCWB	-0.2	0.5	1.2	5.2	13.1	17.3	20.4	19.1	14.1	8.8	3.1	0.9
	2%	DB	-1.1	-0.4	1.5	5.7	16.6	24.0	27.7	21.7	15.4	8.6	2.2	0.9
		MCWB	-1.8	-1.0	0.3	2.5	10.5	16.4	19.1	17.1	12.1	7.3	1.5	0.3
	5%	DB	-2.8	-2.1	0.5	3.8	12.8	21.0	24.9	19.3	13.2	7.0	1.3	0.0
		MCWB	-3.4	-2.7	-0.6	1.7	8.1	14.7	18.2	15.5	10.8	6.2	0.7	-0.7
	10%	DB	-4.6	-3.7	-0.8	2.4	9.5	18.0	22.3	17.1	11.5	5.5	0.5	-1.2
		MCWB	-5.3	-4.4	-1.7	0.9	6.3	13.1	16.5	14.3	9.7	4.7	-0.1	-1.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.0	0.8	1.4	5.3	14.1	19.2	21.9	20.1	15.3	9.2	3.1	1.2
		MCDB	0.5	1.2	2.7	9.5	20.4	26.3	27.6	23.9	17.4	10.7	3.4	1.5
	2%	WB	-1.7	-1.0	0.6	3.1	10.8	17.3	20.3	17.6	12.6	7.4	1.7	0.4
		MCDB	-1.1	-0.5	1.5	5.1	15.7	22.8	25.4	20.8	14.8	8.4	2.0	0.8
	5%	WB	-3.4	-2.7	-0.4	1.9	8.4	15.2	18.9	16.0	11.1	6.2	0.8	-0.7
		MCDB	-2.9	-2.1	0.4	3.3	12.3	19.9	23.7	18.8	12.8	7.0	1.2	-0.1
	10%	WB	-5.2	-4.4	-1.7	1.1	6.3	13.2	17.3	14.5	9.8	4.8	0.0	-1.9
		MCDB	-4.7	-3.7	-0.8	2.3	9.7	17.8	21.5	17.0	11.4	5.5	0.5	-1.2
Mean Daily Temperature Range	5% DB	MDBR	7.4	7.5	8.4	7.7	7.1	7.7	8.4	6.4	5.8	4.2	5.7	6.7
		MCDBR	6.3	6.9	5.0	7.8	12.9	12.9	12.4	10.8	8.8	4.3	4.3	5.1
		MCWBR	6.2	6.9	4.5	5.7	7.7	6.6	6.1	6.4	5.7	3.7	4.4	5.0
	5% WB	MCDBR	6.5	6.9	4.8	6.5	12.2	11.8	11.6	10.0	7.9	4.2	4.2	5.2
		MCWBR	6.4	6.9	4.5	5.0	7.6	6.8	6.4	6.7	5.7	3.8	4.4	5.1
Clear-Sky Solar Irradiance	taub		0.215	0.259	0.300	0.330	0.344	0.348	0.381	0.359	0.324	0.281	0.213	N/A
	taud		1.903	2.033	2.087	2.122	2.293	2.462	2.375	2.423	2.477	2.232	1.942	N/A
	Ebn at Noon		99	531	722	809	847	857	807	786	728	514	91	0
	Edh at Noon		11	52	88	113	107	94	100	85	63	43	9	0
All-Sky Solar Radiation	RadAvg		0.04	0.49	2.01	3.83	4.84	4.93	4.77	3.07	1.71	0.54	0.09	0.00
	RadStd		0.01	0.07	0.20	0.30	0.44	0.51	0.64	0.30	0.18	0.08	0.01	0.00

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page